EDS 223 Homework #2

Geospatial Analysis & Remote Sensing

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Importing Data and Packages

```
library(tidyverse)
library(sf)
library(here)
library(tmap)
library(spData)
library(viridisLite)
library(gt)
```

::: {.cell}

```
# Load EJSreen data
ejscreen <- sf::st_read(here::here("data", "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_CN
```

::: {.cell-output .cell-output-stdout}

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/Users/swaggerboy/EDS223-HW2/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.g
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

:::

```
# Filter EJScreen to Los Angeles County
la <- ejscreen |>
   dplyr::filter(CNTY_NAME == "Los Angeles County")

# Load HOLC Redlining data
holc <- sf::st_read(here::here('data', 'mapping-inequality', 'mapping-inequality-los-angeles
```

::: {.cell-output .cell-output-stdout}

```
Reading layer `mapping-inequality-los-angeles' from data source
  `/Users/swaggerboy/EDS223-HW2/data/mapping-inequality/mapping-inequality-los-angeles.json'
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

:::

```
# Load bike and pedestrian crash reports
crashes <- read.csv('data/SWITRS_LA/Crashes.csv')
```

:::

Part 1: Legacy of Redlining in Current Environmental (In)Justice

```
# Transform la crs to holc crs
holc_trans <- st_transform(holc, crs = st_crs(la))

# Message verification for matching CRS
if(st_crs(holc_trans) == st_crs(la)){
    print("It's a match! :)")
}else{
    warning("It's not a match :(")
}
```

```
[1] "It's a match! :)"
```

Joining holc dataset to la_trans dataset

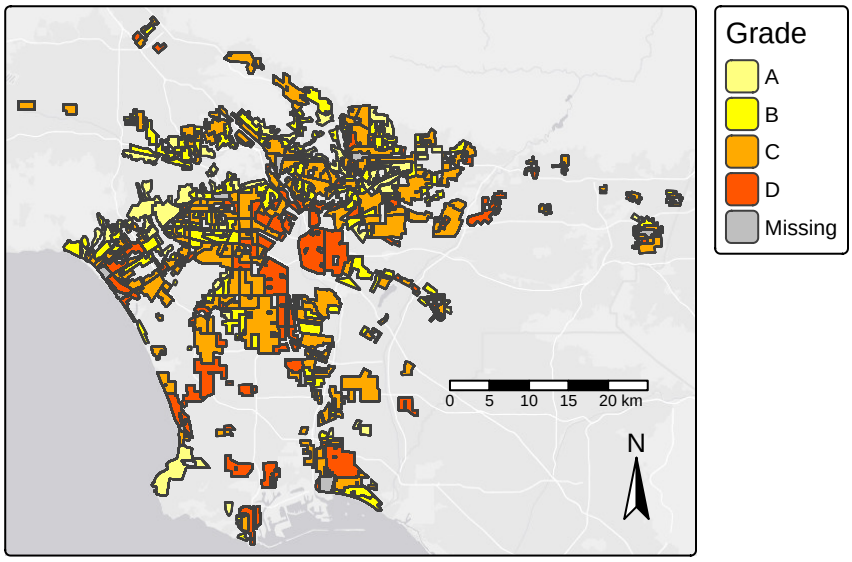
```
# Combine transformed holc data to la county data
combined <- st_join(holc_trans, la)
```

Main Map:

```
# Create map for HOLC grade and census blocks
tm_shape(combined, bbox = holc_trans) +
    tm_polygons(
        fill = 'grade',
        fill.scale = tm_scale(values = rev(heat.colors(5))),
        fill.legend = tm_legend(title = 'Grade')
    ) +
tm_scalebar() +
tm_compass() +
tm_borders() +
tm_title('Grade by Census Block in Los Angeles County') +
tm_basemap()
```

```
[plot mode] fit legend/component: Some legend items or map compoments do not
fit well, and are therefore rescaled.
i Set the tmap option `component.autoscale = FALSE` to disable rescaling.
```

Grade by Census Block in Los Angeles County



Creating Table

```
# Drop geometries from combined
dropped_geom <- st_drop_geometry(combined)

# Filter by grade and create percentage column
filtered <- dropped_geom |>
    group_by(grade) |>
    summarize(
        percentage = n()/nrow(dropped_geom) * 100
    )

# Create table from filtered dataset
filtered |>
    gt() |>
    tab_header(title = md('Percentage of Census Blocks per HOLC Grade'))
```

Percentage of Census Blocks per HOLC Grade

grade	percentage
A	7.028804
B	19.395742
C	47.871008
D	21.070758
NA	4.633688

Means & Visualizations

```
# Create objects for mean values
low_income <- dropped_geom |>
    group_by(grade) |>
    summarize(
        mean_low_income = mean(P_LOWINCPCT, na.rm = TRUE)
    )

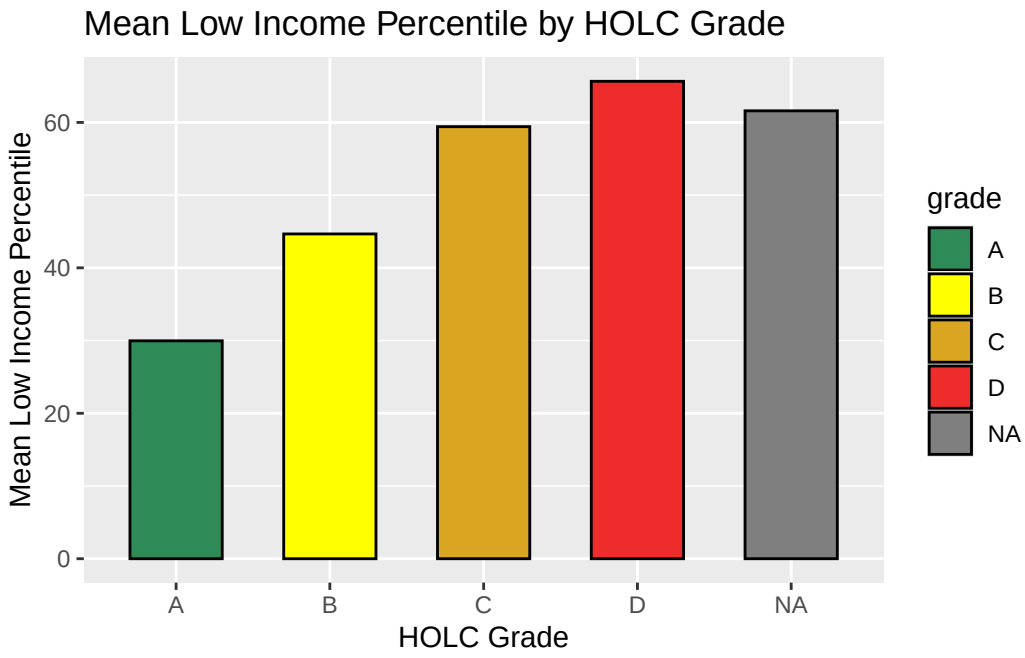
pm25 <- dropped_geom |>
    group_by(grade) |>
    summarize(
        mean_pm25 = mean(P_PM25, na.rm = TRUE)
    )

low_life_exp <- dropped_geom |>
    group_by(grade) |>
    summarize(
        mean_lle = mean(P_LIFEEXPPCT, na.rm = TRUE)
    )
```

Map Displaying Mean Low Income by Grade

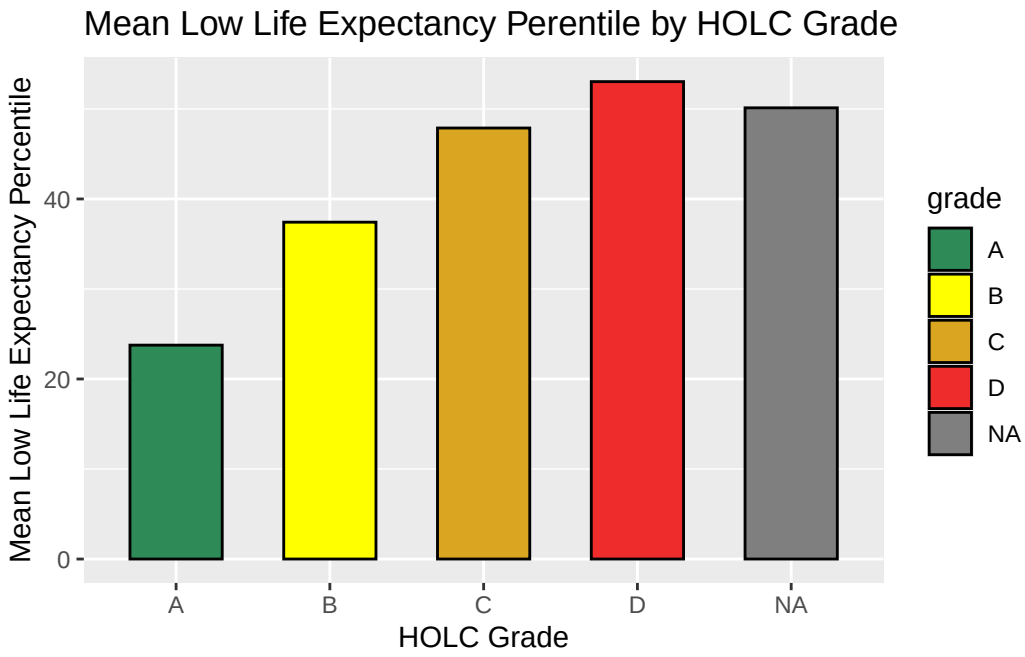
```
# Plot for mean low income by grade
ggplot(
    data = low_income,
    mapping = aes(
        x = grade,
        y = mean_low_income,
```

```
        fill = grade
    )
) +
    geom_col(
        color = 'black',
        width = 0.6
    ) +
        scale_fill_manual(
            values = c(
            'A' = 'seagreen',
            'B' = 'yellow',
            'C' = 'goldenrod',
            'D' = 'firebrick2'
            )
        ) +
        labs(
            title = 'Mean Low Income Percentile by HOLC Grade',
            x = 'HOLC Grade',
            y = 'Mean Low Income Percentile'
        )
```



Map displaying Mean Life Expectancy by Grade

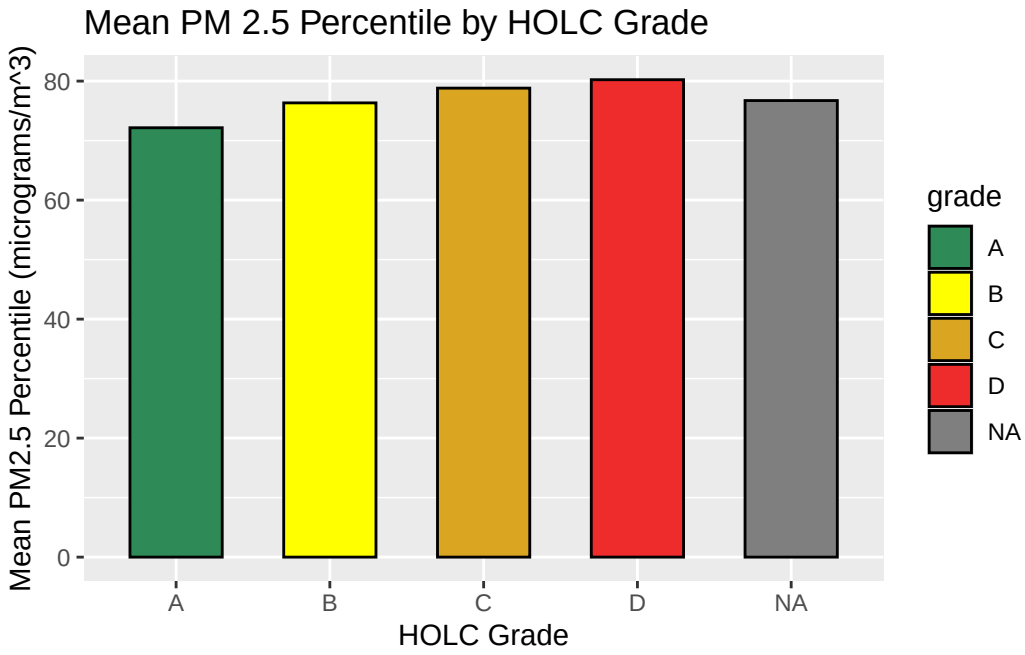
```
# Plot for mean low life expectancy percentile by grade
ggplot(
    data = low_life_exp,
    mapping = aes(
        x = grade,
        y = mean_lle,
        fill = grade
    )
) +
    geom_col(
        color = 'black',
        width = 0.6
    ) +
    scale_fill_manual(
                values = c(
            'A' = 'seagreen',
            'B' = 'yellow',
            'C' = 'goldenrod',
            'D' = 'firebrick2'
                )) +
    labs(
        title = 'Mean Low Life Expectancy Perentile by HOLC Grade',
        x = 'HOLC Grade',
        y = 'Mean Low Life Expectancy Percentile'
    )
```



Map displaying Mean PM 2.5 by Grade

```
# Plot for mean pm 2.5 percentile by grade
ggplot(
    data = pm25,
    mapping = aes(
        x = grade,
        y = mean_pm25,
        fill = grade
    )
) +
    geom_col(
        color = 'black',
        width = 0.6
    ) +
    scale_fill_manual(
        values = c(
            'A' = 'seagreen',
            'B' = 'yellow',
            'C' = 'goldenrod',
            'D' = 'firebrick2')
```

```
  ) +
    labs(
        title = 'Mean PM 2.5 Percentile by HOLC Grade',
        x = 'HOLC Grade',
        y = 'Mean PM2.5 Percentile (micrograms/m^3)'
    )
```



Discussion

Based on the map of redlining in Los Angeles county, the data would suggest that there is a clear relationship between historical redlining and percentage of disadvatanged communities living in areas deemed hazardous (grade D). The visualizations produced displaying the mean variables of interest compared to the concentrations of grading would suggest that higher percentiles of disadvantaged peoples are concentrated in areas that are deemed hazardous. Areas that have the lowest percentage of disadvantaged peoples are deemed the most safest (grade A).

Part 2: Legacy of redlining in pedestrian and bike accidents

Filtering and Transforming Crashes Dataset

```
# only ped and bike accidents
filter <- crashes |>
  filter(PEDESTRIAN_ACCIDENT == "Y" | BICYCLE_ACCIDENT == "Y") |>
  filter(POINT_X != 'NA')

sf_crash <- st_as_sf(filter, coords = c("POINT_X", "POINT_Y"), crs = 4326)

# Transform crash dots to HOLC map CRS
sf_crash <- st_transform(sf_crash, crs = st_crs(combined))
if(st_crs(sf_crash) != st_crs(combined)) stop('Please ensure CRS is matching!')

# Subset points only within HOLC regions
HOLC_regions <- sf_crash[combined, ]

join <- st_join(HOLC_regions, combined) |>
  select(id = ID, region = grade)

# Summarise HOLC grades with most crashes
most_crashes <- join |>
  filter(!is.na(region)) |>
  group_by(region) |>
  summarise(n_points = n()) |>
  st_drop_geometry()

percent <- most_crashes |>
  mutate(Percentage = 100*(n_points/sum(most_crashes$n_points)))
percent$region <- as.factor(percent$region)
percent$Percentage <- as.double(percent$Percentage)

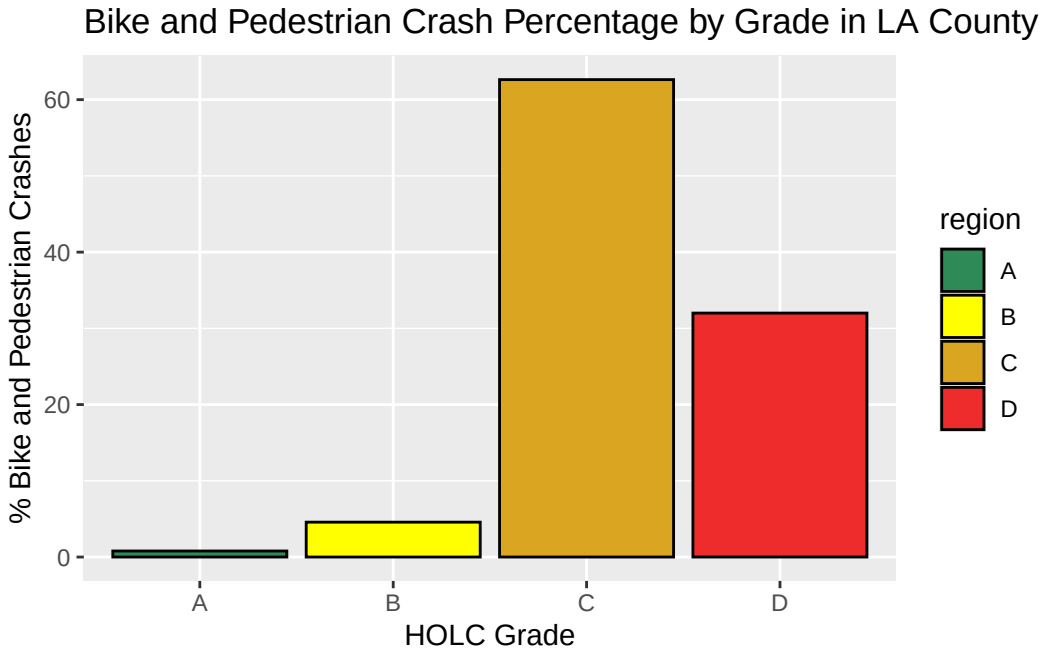
percent
```

```
# A tibble: 4 x 3
  region n_points Percentage
  <fct>     <int>      <dbl>
1 A          4973      0.791
2 B         28788      4.58
3 C        393933     62.6
```

```
4 D         201283     32.0
```

Visualizing Bike and Pedestrian Crashes

```
# graph percentage bike and pedestrian crashes
ggplot(percent,
aes(
    x = region,
    y = Percentage,
    fill = region)
    ) +
  geom_col(
    color = 'black'
  ) +
  scale_fill_manual(
    values = c(
            'A' = 'seagreen',
            'B' = 'yellow',
            'C' = 'goldenrod',
            'D' = 'firebrick2'
            )
    ) +
  labs(x = 'HOLC Grade',
         y = '% Bike and Pedestrian Crashes',
         title = 'Bike and Pedestrian Crash Percentage by Grade in LA County')
```


Bike and Pedestrian Crash Percentage by Grade in LA County
% Bike and Pedestrian Crashes
60
40
20
0
A
B
C
D
HOLC Grade
region
A
B
C
D